

NYC Trip-Duration Serving

Tensium Samples

Task identification

Category

CPU ML engineering

Task name

NYC Trip-Duration Serving

Task slug

cpu-ml-engineering/nyc-trip-duration-serving

Why current models fail, and the capability this task builds

Two of four runs pass; the two failures are the informative ones. Both spent all four rate-limited validation attempts probing the grader instead of reasoning offline about the serving contract, identifier coercion, timezone, and fold-aware DST. The capability this isolates is disciplined offline reasoning under a metering budget: deriving the fix from the data card and worked examples rather than trial-and-error, turning expensive interactive guessing into a cheap one-shot repair.

1. Description

Description

A machine-learning engineering task built around a production trip-duration serving failure. The offline model remains operational, but serving requests lose correct location, fare-code, and local-time context because the feature path mishandles reference joins, mixed identifier serialization, reserved codes, timestamp schema evolution, America/New_York conversion, and daylight-saving boundaries. The agent investigates gated evidence, repairs the serving path, deploys the candidate, diagnoses and recovers an injected schema-registry incident, validates on a hidden real-data slice, promotes, and submits an immutable candidate.

2. Task type & unit type

Task type

Hands-on long-horizon: inspect -> profile -> sample sequential evidence -> infer -> edit -> test -> reproduce -> deploy -> validate -> diagnose -> recover -> validate -> promote -> submit.

Unit type

One isolated Harbor task containing real NYC TLC trips, official lookup tables, an editable pandas/sklearn serving pipeline, a root-owned gated operations service, hidden validation and terminal slices, and a deterministic behavioral grader.

3. Verification method

Verification method

Binary reward by behavior and trajectory. The grader stages the complete submitted candidate environment and runs its serving path as UID/GID 1000 with a timeout. On 6,144 hidden real TLC trips, each of nine categorical outputs must achieve at least 0.9995 agreement with verifier-owned expected behavior, and maximum UNKNOWN-rate drift must not exceed 0.0005. Full reward additionally requires an authenticated ordered gate walk, at least 16 accepted actions and eight observations, correct schema-registry recovery, clean validation, promotion, and submission. Candidate stdout is not a scoring channel.

4. Technical domains & adjacent benchmarks

Technical domains

Classical ML serving; pandas reference joins; nullable and mixed-width identifiers; reserved-code handling; timestamp parsing; timezone conversion; DST ambiguity; offline feature parity; deployment and recovery workflows.

Adjacent benchmarks

Terminal-Bench-like code repair combined with tau-bench-like gated state transitions and hidden deterministic behavioral verification.

5. Human experts used

Human experts

The task was authored, calibrated, and reviewed using the task-authoring and QA process. No additional professional-credential claims are made.

6. Raw base data

Raw base data

Official New York City Taxi and Limousine Commission Green Taxi Trip Records and the official taxi-zone lookup. The visible slice contains 104,003 cleaned trips from 31 December 2023 through 29 February 2024. Hidden validation and terminal slices use official March and November 2024 records around the America/New_York daylight-saving boundaries. The terminal contains 6,144 trips. Source URLs and SHA-256 pins are recorded in the author-side data tooling and visible data card. The records are not simulated or resampled.

Cleaning removes trips with non-positive or greater-than-six-hour elapsed durations and trips with non-positive distance. No target values are synthesized. The maximum measured non-target absolute label correlation is 0.0855.

7. What the world looks like

What the world looks like

The agent starts as an unprivileged Ubuntu user in /workspace/target with editable Python sources, real TLC records, public lookup tables, smoke tests, and an offline backtest. Internet access is disabled. A root-owned Unix-socket service controls observations, deployment, permanently rate-limited validation, recovery, promotion, and submission. Hidden trips, expected terminal behavior, grader code, signing material, and authoritative state remain outside the writable candidate workspace.

8. Time per task

Time per task

Approximately 8–10 authoring and QA hours per task.

9. QA process

QA process

QA runs through a systematic, automated pipeline rather than an ad-hoc pass. Deterministic controls mechanically screen the task for the failure modes that matter, reward-hacking and forgeable graders, answer leakage and reachable oracles, in-process grading, and false difficulty (for example, a "failure" that was only budget truncation). Each control is two-condition validated: it must fire on a planted defect and stay silent on a clean task. An adversarial control battery then executes cheat submissions against the real grader and requires every one to score zero, while a gold solution and a structurally different correct solution both score 1.0. An independent review pass grounds every finding in the actual task files, and difficulty is calibrated by running a frontier model at a fixed sampling standard inside the real isolated environment, with guards that separate genuine failures from budget/truncation artifacts.

Completed TASK-QA-CHECKLIST coverage: gold positive control 1.0; seven structurally different negative/cheat controls 0.0; deterministic repeats; grader isolation; no-new-privileges served-path test; package leak guard; secret scan; and real-model trace review. Two behaviorally different model-produced passing candidates demonstrate that grading is by value rather than source shape.

10. Difficulty calibration

SOTA model & pass@1

Claude Opus 5, using the local claude-personal profile: four independent runs with a maximum of 100 steps each. 2/4 strict passes (50%). Mean reward: 0.5. Passing runs (2, 4) completed the full

gated trajectory in 36-38 steps; run 1 reached the 100-step cap and run 3 called finish() at 86 steps without passing. Infrastructure exceptions: 0. Grader exceptions: 0.

Rollout records

Final local records are supplied separately in nyc-trip-duration-serving_opus5_4traces.zip. The frozen task archive (nyc-trip-duration-serving.zip) SHA-256 is
aa0d29868b4dba8abfc585dbd4f8a6e71ed729a3d5c8832aaadbb1f612cd1387.

11. Grading-design table

Component / gate / axis	What it measures	Weight / cap	How enforced
Behavioral terminal	Exact hidden serving outputs from the deployed candidate , all nine categorical at ≥ 0.9995 agreement, with UNKNOWN-rate drift ≤ 0.0005 .	hard requirement	Candidate executed in an isolated subprocess; hidden scenarios compared by value against verifier-owned expected behavior.
Ordered gate walk	Completion of the required investigation, deploy, recovery and release sequence.	hard requirement	Root-owned authenticated action-ledger replay.
Investigation depth	Genuine long-horizon evidence gathering and operations.	hard requirement	≥ 16 accepted actions and eight accepted observations.
Forced recovery	Correct diagnosis and recovery of the injected schema-registry incident.	hard requirement	Dirty validation -> diagnostics -> schema_registry recovery -> clean validation.
Snapshot binding	The graded code is the submitted code.	hard requirement	No-follow snapshotting, hash binding, immutable submission and full-environment replay.
Isolation / anti-cheat	Candidate cannot read or modify hidden assets or authoritative state.	hard -> 0.0	UID/GID drop, no_new_privs, root-owned assets, signed records, offline execution and tamper controls.

Component / gate / axis	What it measures	Weight / cap	How enforced
Final reward	All behavioral, trajectory, recovery, binding and isolation requirements.	binary 0.0/1.0	Boolean conjunction; partial diagnostics never compensate.

12. How the tasks were created

How created

The task was built from official NYC TLC Green Taxi records and lookup tables after an author-side premise check established real identifier-type drift, reserved codes, missing values, and daylight-saving edge cases. The serving defect combines reference enrichment with timestamp schema evolution and America/New_York behavior. Hidden March and November 2024 slices exercise spring-forward and fall-back boundaries. November contains 71 real records in the repeated 01:00 wall hour; March contains no pickups in the nonexistent 02:00 spring-forward hour.

Fairness is explicit: naive repeated-hour wall times are classified as ambiguous because the source lacks enough information to choose a DST fold; nonexistent spring-forward wall times are classified as nonexistent; offset-bearing timestamps are treated as instants and converted to New York time; ordinary naive timestamps remain local wall time. No secret arbitrary fold is required.

Worked examples use composite records instead of disclosing one repair parameter per example. Zone 265, fractional identifiers, and rate code 99 are withheld from examples but remain inferable from public lookup tables and limited validation. A gold implementation, a structurally distinct merge-based implementation, eight negative controls, Docker isolation checks, a free served hand-drive, and four final Claude Opus 5 runs validated soundness, fairness, and difficulty.

13. QA checklist

- ☒ Two-condition proof: the gold positive control scores 1.0 and wrong solutions score 0.0 for the intended reasons.
- ☒ A second, structurally different merge-based correct solution scores 1.0.
- ☒ Grader, expected outputs, terminal data, backend, and reference solution remain outside /workspace/target.
- ☒ Candidate code runs as an unprivileged subprocess with a timeout and is not imported into the grader.
- ☒ The shipped workspace contains no reference solution or answer key.
- ☒ The required contract is deducible from public tables, source code, composite worked records, and limited validation.

- ☒ Real data is used: official NYC TLC Green Taxi records and official taxi-zone data.
- ☒ Long-horizon behavior is enforced through observations, source repair, deployment, lossy validation, diagnosis, recovery, promotion, and submission.
- ☒ The task ZIP excludes `_private` authoring and calibration material.
- ☒ The package contains no API keys, platform job identifiers, private customer content, or model egress tooling.
- ☒ The workflow is realistic and professionally relevant to production ML serving.
- ☒ Instructions, visible tests, operation gates, and the terminal grader are internally consistent.
- ☒ Dependencies, seeds, hidden slices, hashes, and container execution support deterministic reproduction.
- ☒ The package contains one distinct, self-contained task without duplicate variants.
- ☒ Source material is public, non-sensitive NYC TLC open data; source URLs and hashes are recorded.
- ☒ The package follows the Harbor task structure with instruction, task metadata, environment, tests, and solution directories.

14. Package overview & contents

Field	Submission detail
Selected package	nyc-trip-duration-serving.zip
Task count	1 self-contained Harbor task
Task	cpu-ml-engineering/nyc-trip-duration-serving
Task type	CPU ML engineering; hands-on long-horizon serving investigation, repair, deployment, and recovery
Deliverable	Corrected deployed serving pipeline plus successful validation, promotion, and submission
Verification	Binary behavioral grading over nine outputs, authenticated gate walk, forced recovery, and isolated full-environment replay
Raw base data	Official NYC TLC Green Taxi records and taxi-zone lookup

15. Per-task deep-dive

Task	What the task is about	Deliverable
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NYC Trip-Duration Serving	Diagnose and repair lost location, fare-code, and local-time context in a production trip-duration serving path, including mixed identifiers, reserved records, timestamp schema evolution, and DST behavior.	A corrected deployed pipeline whose nine hidden serving outputs meet the behavioral thresholds, followed by clean validation, promotion, and successful submission.
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16. Example prompt

Our green-taxi duration model is stable offline, but production requests lose location, fare, and local-time context while continuing to return plausible predictions. Work in /workspace/target with no internet. Inspect the editable serving pipeline, real TLC data, public lookup tables, and all sequential evidence returned by the gated operation surface. Infer the correct join, identifier, reserved-code, timestamp, timezone, and DST behavior; repair the serving path; run the visible tests and backtest; then complete reproduce, deploy, rate-limited validation, diagnosis, matching recovery, clean validation, promotion, and submit. Finish only after the corrected candidate is successfully submitted.

17. Environment & execution

Requirement	Specification
Runtime	Python 3.12 slim container; unprivileged ubuntu agent
Dependencies	numpy 2.3.5; pandas 2.3.3; scikit-learn 1.7.2; pyarrow 21.0.0
Compute	4 CPUs; 8,192 MB memory; 10,240 MB storage; 0 GPUs
Time limits	Agent 3,600 seconds; verifier 1,200 seconds; build 3,600 seconds
Visible tests	python3 -m unittest discover -s tests
Backtest	python3 scripts/run_backtest.py
Operations	python3 env_cli.py OP, beginning with inspect, profile, and every requested sample_records cursor
Verification	tests/test.sh runs the root verifier and writes /logs/verifier/reward.json
Network policy	Internet disabled

18. Human, AI & automation roles

Contributor	Role in design and production
Human author & reviewers	Selected and verified the public data, defined the serving contract and fairness rules, reviewed failures, and approved the frozen package.
AI systems	Used for independent calibration and behavioral testing. Four final Claude Opus 5 runs measured difficulty; two independently produced passing candidates completed the task.
Automation	Source downloads and SHA-256 checks, visible tests, container builds, signed action-ledger checks, hidden behavioral probes, privilege-drop checks, alternate-positive execution, and negative controls enforce reproducibility and grading integrity.

19. Supporting artifacts

Artifact	Contents
nyc-trip-duration-serving.zip	Complete clean Harbor task package.
nyc-trip-duration-serving_opus5_4traces.zip	Four independent Claude Opus 5 traces, deployed candidates, grader outputs, rewards, action logs, tool counts, and statuses.
Section 13 of this document	Completed submission checklist.

20. Acknowledgement

The vendor confirms that the task has been reviewed and is realistic, technically substantive, clearly specified and appropriately graded. It reflects a genuine professional workflow and is not a low-effort or artificial task. The creation and calibration statements above are accurate, and the QA checklist has been completed.